

Spencer Lee

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Computational mathematics PhD researcher developing pulse-level control and simulation methods for superconducting qubits to increase gate fidelity, cutting the qubit count and runtime of error correction.

WORK & RESEARCH EXPERIENCE

PhD Researcher

Sep 2021–Present

Michigan State University

East Lansing, MI

- Created and maintain [QuantumGateDesign.jl](#), an open-source Julia package (CI tests, hosted docs) for fast and accurate pulse-level optimal control of superconducting transmon qubits.
- Demonstrated 3–13× faster CNOT gate design than Störmer-Verlet at 10^{-3} – 10^{-5} infidelity targets, using the high-order Hermite methods and discrete adjoints of QuantumGateDesign.jl (*J. Comput. Phys.*, 2026).
- Derived Filon methods that simulate highly oscillatory quantum systems 6× faster than same-order Hermite methods, with no rotating-wave approximation (arXiv:2607.13580; submitted to *SIAM J. Sci. Comput.*).
- Built a transfer-learning pipeline, from dataset generation through model training, that retunes control pulses to preserve gate fidelity on qubits with shifted transition frequencies.
- Presented talks and tutorials at international conferences, including *SIAM Annual Meeting* 2024 & 2025.
- Currently designing neural-network surrogate models for controlled qubit dynamics using learned Magnus expansions, enabling state-vector simulation with large time steps at low computational cost.

Research Project Manager

Jun 2024–Aug 2024

Tohoku University (G-RIPS Program)

Sendai, Japan

- Implemented GPU-accelerated algorithms for classical parameter-setting of the Quantum Approximate Optimization Algorithm (QAOA), in collaboration with Mitsubishi Electric.
- Led a five-member U.S.–Japan team that replaced a state-of-the-art parameter-setting method with a low-cost analytical proxy of comparable accuracy.

Visiting Scholar

Jun 2023–Aug 2023

Los Alamos National Laboratory

Los Alamos, NM

- Wrote CUDA and Thrust code to accelerate simulation of runaway electrons in tokamak fusion reactors.
- Visualized electron dynamics using Poincaré plots to identify computationally demanding reactor regions.

Computing Student Intern

May 2022–Aug 2022

Lawrence Livermore National Laboratory

Livermore, CA

- Calibrated gate pulses and implemented zero-noise extrapolation on real superconducting qubit hardware.
- Measured hardware coherence and accuracy via Ramsey echo and randomized benchmarking experiments.
- Developed high-order compositional integrators for quantum state-vector simulation, which reduced the computational cost of high-accuracy 2-qubit simulations by 10× (*LLNL Technical Report*, 2022).

EDUCATION

PhD in Computational Mathematics and Physics

Oct 2026 (Expected)

Michigan State University

3.94/4.0 GPA

Dual Bachelor of Science in Advanced Mathematics and Physics

May 2020

Michigan State University

High Honors, Honors College, 3.97/4.0 GPA

SKILLS

Languages

Julia, Python, C/C++, VHDL, MATLAB

HPC & Sci. Computing

CUDA, MPI, OpenMP, SLURM, Adjoint Methods, High-Order Methods

Quantum Computing

QuTiP, Qiskit, CUDA-Q, Pulse-Level Control, QAOA, Error Mitigation

Machine Learning / AI

TensorFlow, Scikit-Learn, Lux.jl, Transfer Learning, Scientific ML

Tools

Git, Linux, Tests & CI, Profiling, Agentic AI Workflows, L^AT_EX

AWARDS & HONORS

National Science Foundation Graduate Research Fellowship

May 2022

Michigan State University Rasmussen Fellowship Award

Sep 2021